

Formula SAE

Remote Stop System Information

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RS - Remote Stop System Information

RS.1 Device Specification

RS.1.1 Physical and Electrical Specifications

- RS.1.1.1 The RSS that must be used for the competition is a combination of control box and vehicle module from Gross-Funk GmbH. The specific par number will be listed on the event website.
- RS.1.1.2 All RSS must be of the latest hardware revision (with E-Key) or newer, with consideration to the North American hardware requirement in RS.1.3.1.
- SIL3 (EN61508) certified / EMV certified
 - Signal strength of 25mW
 - 12V to 24V supply voltage (0.26A @12V)
 - 450g, 173mm × 113mm × 35mm
 - IP20 (receiver, Vehicle Module) / IP65 (sender, Control Box)
 - Refer to the GrossFunk documentation on the event website for ordering information.
- RS.1.1.3 The vehicle module includes a normally-open (NO) relay which must be part of the shutdown circuit. It opens on switching shutdown, on signal loss, and on power loss. The maximum current rating is 4A.



Figure 1. RSS Control Box and Vehicle Module (North American device colors may differ)

RS.1.2 Communication Specification

RS.1.2.1 The CANopen interface of the vehicle module has the following properties:

- 1000 kbit/s, 125 kbit/s, 250 kbit/s and 500 kbit/s in standard configuration.
- Cyclic PDOs containing states of switches (Go-signal) and radio
- Warns if signal loss detected (200ms in advance to shutdown, contained in cyclic PDO)

RS.1.2.2 The Node-ID must be set to 0x011 and baud rate set to 500kbps during the competition. The Node-ID and baud rate settings of the vehicle module can be configured with the external DIP switches.

DIP Switch	1	2	3	4	5	6	7	8	Baud Rate
Node ID	+1	+2	+4	+8	+16	+32	0	0	1Mbps
Bit	0	1	2	3	4	5	1	0	125kbps
							0	1	250kbps
							1	1	500kbps
	Node ID						Baud Rate		

Table 1. RSS DIP switch configuration

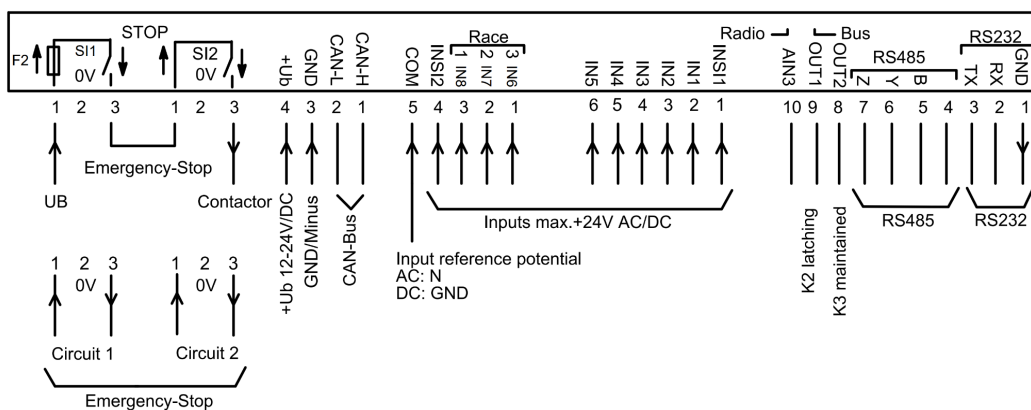


Figure 2. RSS Vehicle Module Connections

RS.1.2.3 When the vehicle module is booted up it will send a message to signalize its initialization (NMT message with CAN-ID 0x700 + Node-ID and a single data byte 0x00). A CAN/CANopen master device must set the receiver to operational mode (NMT message CAN-ID = 0x000, byte 0 = 0x01 (requested state), byte 1 = addressed Node-ID or 0x00 for all). After setting the vehicle module to operational mode, the vehicle module starts sending a status message of 8 bytes containing PDOs 2000 - 2007 (one byte each, CAN-ID = 0x180 + Node-ID) every 30 ms.

RS.1.2.4 Manually resetting the RSS before sending the operational mode message may be used to check if the device is online (NMT message CAN-ID = 0x000, byte 1 = 0x80 (requested state), byte 2 = addressed Node-ID). This will be answered with the boot-up message.

- RS.1.2.5 Beside the CAN-IDs mentioned in RS.1.2.3 and RS1.2.4, be aware not to use the CANopen-related IDs listed in Table 2 on the bus.

Communication Object	CAN ID	Receive Nodes
NMT Node Control	0x000	Receive Only
Sync	0x080	Receive Only
Emergency	0x080 + Node ID	Transmit
Timestamp	0x100	Receive Only
PDO	0x180 + Node ID 0x200 + Node ID	Transmit PDO Receive PDO
SDO	0x580 + Node ID 0x600 + Node ID	Transmit Receive
NMT Node Monitoring	0x700 + Node ID	Transmit
LSS	0x7E4 0x7E5	Transmit Receive
Table 2. Reserved Message IDs		

- RS.1.2.6 System misbehavior and faulty logs caused by misuse of these messages eliminates the demand for a re-run and may lead to a disqualification. This penalty may also be enforced for any kind of hardware manipulation to the control box or vehicle module, or improper antenna modifications..
- RS.1.2.7 The status of the switch (K2) and the button (K3) at the control box is contained in the PDO 2000 (bit 1 and 2) as well as on the digital outputs, see Figure 7. The E-Stop is signaled by PDO 2000 bit 0 and PDO 2003 bit 7. PDO 2006 contains the radio quality (0% to 100%) whereas PDO 2007 summarizes several radio states, i.e. the pre-alarm radio communication interruption (bit 6, 200 ms in advance to shutdown).
- RS.1.2.8 Either K2 or K3 are allowed to be used to signalize the Go-signal for switching from “Ready” to “Driving” state, see DO.1.1.1. Both the CAN message or the digital outs can be used.

RS.1.3 Control Box Keys

- RS.1.3.1 The North American competition may require teams to purchase a North America specific E-key to use in their remote stop equipment. Refer to the Event Website for updates on this requirement.
- RS.1.3.2 Teams may train using the training E-Key they received when purchasing their equipment.
- RS.1.3.3 For dynamic events, the officials will hand-out a Race E-Key that has to replace the team’s Training E-Key for the time of the run (switching the RSS to a different set of frequencies

within the allowed range). The “Race Key” must be returned to the officials immediately after the run has been finished.

- RS.1.3.4 In order to enable the Race E-Key frequencies at the vehicle module, the input “Race 1” has to be set to high (by bridging the input with supply “+Ub”). That needs to be done upon receipt of the E-Key with a flip switch in proximity to the Driverless Mission Indicator (DMI). Race mode position has to be marked with an “R”. Correct mode selection can be traced via the input’s LED as well as in PDO 2007, bit 7.
- RS.1.3.5 Teams may attempt their EBS test runs with the Race Key to verify functionality up until the opening of the first dynamic event. After which the team must continue to use their training key.

RS.2 Data Logging

RS.2.1 Energy Meter Signals

- RS.2.1.1 The Remote Stop System must be connected to the FSAE Energy Meter’s CAN bus with a baud rate according to RS.1.2.2.
- RS.2.1.2 Beside Remote Stop System (RSS) messages mentioned in RS.1.2.3 and RS.1.2.5, the messages defined in Table 3 may be provided to the FSAE Energy Meter with a cycle time of 100ms, according to RS.2.1.4.
- RS.2.1.3 All Driverless signals are little-endian (Intel). Scale, if not defined, is 1.
- RS.2.1.4 All pressure signals mentioned in the team’s Driverless System Form (DSF) should be provided in message with CAN-ID 0x500. If there are more than 4 signals available, make sure at least one full redundant path is represented.
- RS.2.1.5 A valid dbc file containing the message definition of the DSF messages must be uploaded as part of the DSF submission.

ID	0x500 (11-bit)										
Name	EBS Pressure Status										
	Bits	7	6	5	4	3	2	1	0	Unit	Scaling
Bytes	0	ebs_pressure_1 (int16)								bar	1/16
	1										
	2	ebs_pressure_2 (int16)								bar	1/16
	3										
	4	ebs_pressure_3 (int16)								bar	1/16
	5										
	6	ebs_pressure_4 (int16)								bar	1/16
	7										

Table 3. Message definition of optional logged EBS pressure data